

BUSINESS ANALYTICS

Group Project Announcement

Learner Behavior Segmentation for an Online Learning Platform

An End-to-End Analytics Project: From Raw Data to a Monitored, Deployed Model

Course	Business Analytics
Project Type	Group Project (4–5 students per team)
Domain	EdTech — Learner Engagement & Success
Deployment Target	Cloud free tier (e.g., Render, Hugging Face Spaces, Streamlit Community Cloud)
Coding Requirement	All code must be written by team members (no no-code/AutoML platforms)

1. Project Overview

Online learning platforms generate rich behavioral data — video views, quiz attempts, forum posts, login patterns — yet most platforms still treat every learner the same way. Identifying distinct learner profiles allows a platform to personalize content pacing, target at-risk learners with timely interventions, and design better course experiences. This project asks each team to act as the learner-insights analytics function of an online learning platform and answer a concrete business question:

“What distinct behavioral profiles exist among our learners, and how should content, pacing, or support be tailored to each group?”

Teams will build a clustering model that groups learners by engagement and performance behavior, then translate each cluster into an actionable, named persona (e.g., “consistent achiever”, “last-minute crammer”, “at-risk disengaged”). The model will be engineered, deployed as a working application, and monitored after deployment — mirroring the full lifecycle of an analytics product in industry.

2. Data Sources

Teams must combine the following two data sources:

- **Kaggle dataset (required base):** “Open University Learning Analytics Dataset (OULAD)” — [kaggle.com/datasets/anlgrbz/student-demographics-online-education-dataoulad](https://www.kaggle.com/datasets/anlgrbz/student-demographics-online-education-dataoulad). This provides learner activity logs and assessment records.

- **Synthetic data (required extension):** Each team must generate additional behavioral data using Python (e.g., the Faker library plus custom business logic) — login frequency and time-of-day pattern, video completion rate, days between enrollment and first activity, forum/discussion participation, and assignment submission timeliness.

This combination ensures the project is not a simple “download-and-train” exercise: teams must justify the realism of the behavioral features they generate and document the simulation logic rigorously.

Data Dictionary Requirement: Every synthetically generated field must be documented in a data dictionary specifying column name, data type, unit, valid range, and the generation logic or business assumption used.

3. Project Structure (8 Modules)

The project is organized into eight modules, suitable for an 8-week timeline (adjustable to fit the course schedule).

Module	Focus Area	Key Tasks
1	Business Understanding & Data Generation	• Define the business problem and key KPIs (engagement score, at-risk learner rate, intervention cost) • Generate supplementary behavioral/activity data in Python • Produce a complete data dictionary
2	Exploratory Data Analysis (EDA)	• Analyze distributions, outliers, and correlations among engagement and performance variables • Compare behavior patterns across course type, cohort, and enrollment period • Visualize relationships (e.g., login frequency vs. final score, video completion vs. submission timeliness)
3	Data Cleaning	• Handle missing values, duplicates, and inconsistent categorical labels • Treat outliers (e.g., bot-like login patterns, implausible activity counts) • Document all cleaning decisions with before/after comparisons
4	Feature Engineering	• Construct behavioral features (RFM-style: recency of last login, frequency of activity, engagement intensity) • Encode categorical variables, scale numerical features, and reduce dimensionality if needed (e.g., PCA) • Select the feature set most relevant to distinguishing learner behavior
5	Model Development	• Train and compare clustering algorithms: K-Means, Hierarchical Clustering, and DBSCAN

Module	Focus Area	Key Tasks
		- Determine the optimal number of clusters (elbow method, silhouette score) and validate cluster stability - Profile each cluster, assign an interpretable persona name, and characterize what distinguishes each group
6	Model Deployment	- Package the model as an API (FastAPI or Flask) that assigns a new learner to a segment based on their activity profile - Build a demo dashboard (Streamlit) showing cluster profiles and allowing instructors to look up a learner's segment - Deploy to a cloud free tier (Render, Hugging Face Spaces, or equivalent)
7	Model Monitoring	- Track data drift by comparing incoming learner-activity distributions to training data - Monitor cluster stability over time (e.g., whether segment sizes or centroids shift significantly) - Build a simple monitoring dashboard (e.g., using the open-source Evidently library) and define re-clustering triggers
8	Final Report & Presentation	- Translate cluster profiles into pedagogical and support recommendations for each learner segment - Discuss limitations and future improvements - Present to the class with a live or recorded demo

4. Deliverables

Deliverable	Format
Source code (notebooks/scripts)	GitHub repository
Data dictionary	Excel or Markdown file
Deployed API + segment lookup dashboard demo	Live link (Render / Hugging Face Spaces / Streamlit Cloud)
Monitoring dashboard or report	Notebook or Evidently report
Final report and slides	PDF and PPTX

5. Suggested Team Roles (4–5 members)

Role	Responsibilities
Data Engineer	Synthetic behavioral data generation, data dictionary, data cleaning

Role		Responsibilities
Data Analyst		Exploratory data analysis and visualization of learner behavior
ML Engineer Modeling	—	Feature engineering and clustering model development
ML Engineer Evaluation	—	Cluster-quality validation (silhouette score, stability checks) and persona definition
MLOps Lead		Deployment and monitoring (may be combined with another role for 4-person teams)

6. Evaluation Focus

Grading will emphasize the rigor and clarity of each stage, rather than model accuracy alone:

- Quality and realism of synthetic behavioral data, supported by a complete data dictionary
- Depth of EDA and clarity of insights communicated through visualization
- Soundness of cleaning and feature engineering decisions, with clear documentation
- Rigor in selecting the number of clusters and validating cluster quality (e.g., silhouette score)
- Clarity and business usefulness of the persona names and profiles assigned to each cluster
- A working, accessible deployment (live link required)
- A functioning monitoring component with defined drift/re-clustering triggers
- Originality of code — all code must be authored by team members

7. Submission Logistics

Instructors should insert specific dates for each milestone below before distributing this announcement.

1. Team formation and topic confirmation — [insert date]
2. Module 1–3 checkpoint (data + EDA + cleaning) — [insert date]
3. Module 4–5 checkpoint (features + models) — [insert date]
4. Module 6–7 checkpoint (deployment + monitoring) — [insert date]
5. Final submission and presentation — [insert date]

Questions about scope, data sources, or deployment options should be directed to the course instructor.